

# GAUGE-ALIGNED ENDPOINTS AND CROSS-DIMENSIONAL TOPOLOGY FOR TEXT EMBEDDING DISTILLATION

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Paper under double-blind review

## ABSTRACT

Embedding distillation transfers knowledge from a large teacher to a smaller student, yet their representations often differ in dimension and orientation. Existing methods handle this mismatch with trainable projections, intermediate-layer supervision, or auxiliary objectives. We propose GATE-KD, a final-embedding framework centered on a comparable endpoint interface. A fixed principal-component projection selects a teacher subspace at the student dimension, while closed-form orthogonal alignment follows the evolving student across epochs without altering the projected teacher geometry. A complementary topology loss matches connectivity directly between the original teacher and student spaces despite their different dimensions. GATE-KD uses only cached final teacher embeddings and introduces no trainable alignment module. Across three teacher-student configurations and nine downstream benchmarks, GATE-KD obtains the highest aggregate score among the evaluated distilled students, improving over the next-best baseline by 0.38–1.59 points while reducing measured GPU optimization time by 62–74%. Controlled analyses show that alignment matters beyond retained teacher variance and that native-space connectivity complements point-wise endpoint supervision. These results establish a carefully designed endpoint interface as a competitive alternative to more complex distillation pipelines.

## 1 INTRODUCTION

Text embedding models support retrieval, semantic similarity, clustering, and classification, but the strongest encoders are often too costly for latency- or memory-constrained deployment. Knowledge distillation offers a natural route to compact embedding models by transferring supervision from a frozen teacher to a smaller student (Hinton et al., 2015; Reimers & Gurevych, 2020). Unlike class logits, however, embedding coordinates have no shared semantics across models. Teacher and student representations may differ in dimension, orientation, depth, and even architecture. Existing methods address this mismatch with learned projectors, relational or contrastive objectives, and supervision across intermediate layers (Park et al., 2019; Tian et al., 2020; Xu et al., 2023; Zhang et al., 2024a; Truong et al., 2025; Dao et al., 2026). Intermediate alignment can provide rich supervision, but it presupposes useful correspondences between teacher and student depths—an assumption that becomes fragile as their architectures diverge—and often requires additional mappings or repeated access to teacher hidden states (Ko et al., 2023; Yu et al., 2025). More fundamentally, this machinery can obscure whether the difficulty lies in coordinating the representation hierarchy or in failing to define a comparable final representation in the first place. This motivates our hypothesis that the endpoint interface is an under-examined bottleneck in cross-dimensional distillation.

We test this hypothesis by separating teacher knowledge into two channels. The *extrinsic interface* describes how teacher representations are expressed in coordinates that the student can match point-wise, whereas *intrinsic structure* comprises properties of the representation cloud that do not depend on its coordinates or ambient dimension. This decomposition leads to **GATE-KD**, Gauge Alignment and Topology for Embedding Distillation. GATE-KD fits a fixed principal-component subspace to cached teacher endpoints, selecting the information that can pass through the student-width bottleneck. Within that subspace, an orthogonal Procrustes update aligns the target coordinates to the current student and is refitted after each epoch on a fixed calibration subset. Because each update

is orthogonal, it changes only the targets’ coordinate gauge, leaving their pairwise geometry unchanged. In parallel, GATE-KD matches the degree-zero persistent-homology signatures of student batches and the original, unprojected teacher batches. This second term transfers multiscale connectivity directly between native spaces of different dimensions. The resulting objective combines pointwise endpoint correspondence with coordinate-invariant structure while supervising only the final student embedding. It trains no projector, uses no intermediate-layer objective, and requires no online teacher forward pass during ordinary training.

Experiments across three teacher–student configurations and nine downstream benchmarks support this endpoint-centered view. Under endpoint-only supervision, adding the Procrustes-aligned gauge to the same PCA subspace raises the average score from 69.35 to 74.28, showing that retained teacher variance alone does not determine whether a target is easy to learn. Native-space  $H_0$  supervision then improves the endpoint-only result from 74.28 to 74.79, indicating that intrinsic connectivity provides a complementary refinement. With both signals active, refitting the gauge across epochs adds 0.31 points over retaining the alignment fitted at initialization. Overall, GATE-KD achieves the highest aggregate score among the evaluated distilled students in all three configurations, exceeding TALAS by 0.38, 0.48, and 1.59 points while using only 26–38% of its measured GPU optimization time. These results show that, in the settings we study, strong embedding transfer does not require coordinating the intermediate representation hierarchy when the endpoint interface and native-space structure are specified explicitly.

We make three contributions.

- We formulate cross-dimensional embedding distillation as an intrinsic–extrinsic decomposition, separating the coordinate interface required for pointwise supervision from structure that can be transferred directly across native spaces.
- We introduce GATE-KD, a terminal-only method that combines a fixed PCA subspace, epoch-wise closed-form gauge alignment, and native-space  $H_0$  persistence matching using only cached final teacher embeddings and no trainable alignment module.
- Through controlled interface and signal decompositions, we show that coordinate comparability and topology provide complementary supervision; across three teacher–student configurations, this simple formulation improves aggregate performance while substantially reducing measured optimization cost relative to a strong hierarchy-based baseline.

## 2 RELATED WORK

Early text-embedding distillation directly regressed student sentence vectors onto teacher outputs, showing that unlabeled text can provide dense supervision without task annotations (Reimers & Gurevych, 2020). Subsequent methods have enriched this signal through contrastive learning, specialized training recipes, and teacher-compatible output spaces (Xu et al., 2023; Zhang et al., 2024a; Li et al., 2025; Vujanic & Rückstieß, 2026). Recent systems also emphasize the data and task recipe: Gecko distills LLM-generated and relabeled retrieval data, while jina-embeddings-v5-text combines embedding distillation with task-targeted contrastive training (Lee et al., 2024; Akram et al., 2026). These systems demonstrate the practical value of compact embedding models but jointly change the transfer data and objective. Our study instead holds the unlabeled corpus and downstream protocol fixed to isolate the representation interface between a given teacher and student. EmbedDistill combines embedding matching with relative geometric supervision for information retrieval (Kim et al., 2023). Other approaches draw on the representation hierarchy. EMO transfers token-level relations and aligns hidden states across tokenizers, while TALAS anchors upper student layers to the teacher endpoint and propagates supervision through lower layers by self-distillation (Truong et al., 2025; Dao et al., 2026). MoL replaces rigid layer pairing with similarity-weighted mixtures of teacher layers, and feature-alignment distillation likewise transfers intermediate states and attention (Vu et al., 2026; Wang et al., 2024). Intermediate supervision can provide additional signals, but it also introduces choices about layer correspondence and often requires teacher hidden states or auxiliary mappings. Prior analyses have identified overfitting in intermediate-layer distillation and found that its layer-selection strategy may have limited influence (Ko et al., 2023; Yu et al., 2025). GATE-KD studies the complementary regime in which the teacher is accessed only through cached final embeddings and every training signal is applied to the student endpoint.

Dimension mismatch in feature distillation is commonly handled by mapping one representation space into the other. HPD learns projection layers that produce compact sentence representations (Zhao et al., 2022), and broader studies show that a projector’s parameterization and normalization can substantially change the student’s optimization dynamics (Miles & Mikolajczyk, 2024; Chen et al., 2023). Orthogonally constrained projections preserve intra-batch similarity, while TCS uses PCA coordinates extracted from cached teacher features (Miles et al., 2024; Zhou et al., 2025). More generally, representation-similarity work treats orthogonal transformations as natural equivalences because they preserve inner products and distances, with Procrustes providing the corresponding alignment (Kornblith et al., 2019; Williams et al., 2021; Maystre et al., 2025). Recent work formalizes teacher supervision over representation equivalence classes (Han, 2026). GATE-KD instantiates this perspective for cross-dimensional text embeddings through a terminal-only interface combined with native-space  $H_0$  transfer. Most closely related to our geometric motivation, Bhattarai et al. (2025) show that unconstrained projected matching can attain zero feature loss without preserving the teacher’s Gram geometry. Under their assumptions, zero Procrustes distance is instead equivalent to zero Gram discrepancy. GATE-KD instead uses Procrustes to estimate a coordinate gauge on a fixed calibration set. The gauge is held constant within each epoch and updated in closed form as the student evolves. PCA selects the retained teacher subspace, while the orthogonal update changes its orientation without altering the projected teacher geometry.

Coordinate-free objectives offer another route to transferring representation structure. Relational distillation matches pairwise distances or angles, similarity-preserving distillation matches Gram structure, and contrastive distillation transfers relations through positive and negative examples (Park et al., 2019; Tung & Mori, 2019; Tian et al., 2020). Persistent homology extends this view from pairwise relations to multiscale properties of a point cloud. In graph representation distillation, G-CRD studies preservation of local and global embedding topology through contrastive alignment (Joshi et al., 2024). Topology-related embedding objectives also appear in more specialized settings: JD-CEmb combines relational distillation with contrastive learning for task-oriented dialogue, whereas DICE modifies text-conditioning embeddings to absorb classifier-free guidance in diffusion models (Burykina et al., 2025; Zhou et al., 2026). The latter settings motivate geometry-aware embedding objectives but do not address cross-dimensional sentence-encoder compression. TopKD transfers global feature topology through approximate persistence images, while topological-guidance distillation combines image-derived persistence features from multiple teachers (Kim et al., 2024; Jeon et al., 2024). GATE-KD uses topology neither as a surrogate for all representation geometry nor as a replacement for sample-wise supervision. It directly matches the sorted degree-zero persistence death times of student batches and unprojected teacher batches, which requires neither shared coordinates nor equal ambient dimension. The resulting decomposition places coordinate-dependent correspondence behind an explicit endpoint interface while allowing coordinate-invariant connectivity to transfer directly across native spaces.

### 3 METHOD

Let  $f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}$  be a frozen teacher and  $f_S : \mathcal{X} \rightarrow \mathbb{R}^{d_S}$  a student with  $d_T > d_S$ . We run the teacher once over an unlabeled corpus  $\{x_i\}_{i=1}^N$  and cache its normalized final sentence embeddings

$$\mathbf{t}_i = \text{norm}(f_T(x_i)) \in \mathbb{S}^{d_T-1}, \quad (1)$$

where  $\text{norm}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$ . The corresponding student embedding is  $\mathbf{z}_i = \text{norm}(\text{Pool}(h_i^{(L)})) \in \mathbb{S}^{d_S-1}$ . GATE-KD transfers teacher knowledge through two terminal channels. An extrinsic interface expresses each teacher endpoint in coordinates that the student can match pointwise, while an intrinsic objective transfers structure that does not require shared coordinates or equal ambient dimension.

#### 3.1 EXTRINSIC ENDPOINT INTERFACE

**Subspace selection.** Let  $\mu = N^{-1} \sum_i \mathbf{t}_i$ . We fit the leading  $d_S$  principal directions of the centered cache and collect them as  $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$ , with  $P_{\text{PCA}}^\top P_{\text{PCA}} = I_{d_S}$ . Centering is used to estimate the principal directions; in the evaluated recipe, the cached vectors are not centered when the projection is applied. The normalized coordinates before gauge alignment are

$$\mathbf{y}_i = \text{norm}(\mathbf{t}_i P_{\text{PCA}}) \in \mathbb{S}^{d_S-1}. \quad (2)$$

The PCA subspace is fitted once on the full teacher cache and remains fixed throughout training.

**Gauge alignment.** The basis used to express the retained subspace affects a pointwise loss even though it does not affect the geometry within that subspace. We resolve this coordinate ambiguity with orthogonal Procrustes alignment (Schönemann, 1966; Bhattarai et al., 2025). A calibration subset  $\mathcal{C}$  is selected before training and reused unchanged. Let  $Y_{\mathcal{C}}$  contain its projected teacher coordinates and let  $Z_{\mathcal{C}}^{(e)}$  contain the current student embeddings at the start of epoch  $e$ . We compute

$$R^{(e)} = \arg \min_{R \in O(d_S)} \|Y_{\mathcal{C}} R - Z_{\mathcal{C}}^{(e)}\|_F^2 = UV^\top, \quad U\Sigma V^\top = \text{svd}(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)}). \quad (3)$$

The initialized student determines  $R^{(0)}$ . After each non-final epoch, we re-encode the same calibration sentences and refit the gauge for the following epoch. The target used within epoch  $e$  is therefore

$$\tau_i^{(e)} = y_i R^{(e)}. \quad (4)$$

Each target is fixed within an epoch. The refit is a closed-form coordinate update rather than a learned projector.

**Relation to the endpoint objective.** Because the rows of  $Y_{\mathcal{C}}$  and  $Z_{\mathcal{C}}^{(e)}$  have unit norm, their cosine endpoint objective satisfies

$$\frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \left(1 - \langle \mathbf{z}_i^{(e)}, y_i R \rangle\right) = \frac{1}{2|\mathcal{C}|} \|Z_{\mathcal{C}}^{(e)} - Y_{\mathcal{C}} R\|_F^2. \quad (5)$$

Equation 3 therefore exactly minimizes the calibration endpoint loss over  $R$  at each refit boundary with the student fixed.

**Gauge invariance.** For every orthogonal  $R$ ,

$$(YR)(YR)^\top = YY^\top. \quad (6)$$

Consequently, refitting the gauge leaves all pairwise cosines, chord distances, and persistent-homology signatures of the projected teacher targets unchanged. PCA determines which teacher subspace survives the width bottleneck, while Procrustes determines how that fixed geometry is presented to the evolving student.

**Corollary (Monotone gauge refit).** *With the student fixed, every gauge refit weakly decreases the calibration endpoint loss, while leaving the topology term unchanged.*

*Proof.* Equation 3 minimizes the endpoint loss over all  $R \in O(d_S)$ , so its value cannot exceed that obtained with the previous gauge. The topology term depends only on the fixed native teacher and student embeddings and is independent of  $R$ .  $\square$

### 3.2 INTRINSIC TOPOLOGY TRANSFER

The endpoint interface can reproduce only the geometry retained after projection. Although  $P_{\text{PCA}}$  has orthonormal columns,  $P_{\text{PCA}} P_{\text{PCA}}^\top$  is a rank- $d_S$  projector, so the cosine Gram matrix of the normalized coordinates in Equation 2 need not equal that of the original teacher embeddings. We therefore transfer an additional structural signal directly from the teacher’s native space.

This loss of geometry is unavoidable whenever the normalized teacher cache has rank above the student width.

**Proposition 1 (Irreducible Gram distortion).** *Let  $T \in \mathbb{R}^{N \times d_T}$  collect the normalized teacher cache embeddings, with singular values  $\sigma_1(T) \geq \dots \geq \sigma_r(T) > 0$ . For every  $Z \in \mathbb{R}^{N \times d_S}$  collecting student embeddings of the same  $N$  sentences,*

$$\|TT^\top - ZZ^\top\|_F^2 \geq \sum_{j > d_S} \sigma_j(T)^4. \quad (7)$$

*Thus exact preservation of the teacher cosine Gram matrix is impossible when  $\text{rank}(T) > d_S$ .*

*Proof.* The eigenvalues of  $TT^\top$  are  $\sigma_j(T)^2$ , whereas  $\text{rank}(ZZ^\top) \leq d_S$ . The Eckart–Young theorem therefore lower-bounds the squared error of any such rank- $d_S$  approximation by the squared tail eigenvalues (Eckart & Young, 1936).  $\square$

For a minibatch of  $B$  normalized vectors, we use the chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}. \quad (8)$$

The  $B - 1$  finite death times of degree-zero Vietoris–Rips persistence are the edge weights of the point cloud’s minimum spanning tree (Hofer et al., 2020). Let  $\delta_{(1)}^S \leq \dots \leq \delta_{(B-1)}^S$  and  $\delta_{(1)}^T \leq \dots \leq \delta_{(B-1)}^T$  denote the sorted death times of the student embeddings  $\{\mathbf{z}_i\}$  and the original teacher embeddings  $\{\mathbf{t}_i\}$ . We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left( \delta_{(j)}^S - \delta_{(j)}^T \right)^2. \quad (9)$$

For a precise interpretation, define the empirical distributions of finite connectivity scales

$$\nu_S = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{\delta_{(j)}^S}, \quad \nu_T = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{\delta_{(j)}^T}, \quad (10)$$

where  $\delta_a$  denotes a Dirac mass at  $a$ .

**Proposition 2** (Transport interpretation and stability). *The  $H_0$  loss equals the squared one-dimensional Wasserstein distance between the empirical death-scale measures,*

$$\mathcal{L}_{H_0} = W_2^2(\nu_S, \nu_T). \quad (11)$$

Moreover, if the two distance matrices on the corresponding batch satisfy

$$\epsilon = \max_{i,j} |d(\mathbf{z}_i, \mathbf{z}_j) - d(\mathbf{t}_i, \mathbf{t}_j)|, \quad (12)$$

then  $\mathcal{L}_{H_0} \leq \epsilon^2$ .

*Proof.* Optimal transport on the real line pairs equally weighted samples in sorted order, which gives Equation 11 (Peyré & Cuturi, 2019). Under a uniform edge perturbation of at most  $\epsilon$ , every threshold graph at scale  $a$  in either metric is contained in the other at scale  $a + \epsilon$ . Kruskal’s characterization of the ordered minimum-spanning-tree edges then gives  $|\delta_{(j)}^S - \delta_{(j)}^T| \leq \epsilon$  for every  $j$ , and averaging their squared differences proves the bound. This is the degree-zero finite-space case of Vietoris–Rips persistence stability (Chazal et al., 2014).  $\square$

The teacher diagram is computed from  $\mathbf{t}_i \in \mathbb{R}^{d_T}$  rather than  $\mathbf{y}_i$  or  $\tau_i^{(e)}$ . The loss therefore needs neither a projection nor a shared orientation. It is piecewise differentiable through the selected student edge lengths (Hofer et al., 2019). The stability result is one-way. Equation 9 matches the multiset of  $H_0$  death times but does not impose correspondence between individual minimum-spanning-tree edges, recover the full metric geometry, or preserve higher-dimensional topological features.

### 3.3 OBJECTIVE AND ALTERNATING OPTIMIZATION

The endpoint loss for a training minibatch is

$$\mathcal{L}_{\text{end}}^{(e)} = \frac{1}{B} \sum_{i=1}^B \left( 1 - \langle \mathbf{z}_i, \tau_i^{(e)} \rangle \right). \quad (13)$$

GATE-KD optimizes the two-term objective

$$\mathcal{L}^{(e)} = \mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \quad (14)$$

Here,  $\mathcal{L}_{H_0}$  is evaluated on the same minibatch using the student embeddings and the corresponding unprojected teacher embeddings.

Optimization alternates between stochastic updates of the student parameters and exact updates of the coordinate gauge. During epoch  $e$ ,  $R^{(e)}$  is held fixed while the student is optimized with Equation 14. After epoch  $e$ , unless it is the final epoch, the updated student re-encodes  $\mathcal{C}$  and Equation 3 produces  $R^{(e+1)}$  for the next epoch. The  $H_0$  term is unaffected by these gauge updates because it is computed from the native teacher embeddings.

Teacher embeddings and the PCA subspace are computed once before student optimization. Ordinary training steps use the cached teacher embeddings and require no online teacher forward pass. Each gauge update adds one student pass over the calibration set and an SVD of a  $d_S \times d_S$  matrix.

## 4 EXPERIMENTS

### 4.1 EXPERIMENTAL SETUP

**Teacher–student configurations.** We adopt the three teacher–student pairs evaluated by TALAS (Dao et al., 2026). They comprise Qwen3-Embedding-4B  $\rightarrow$  BERT-base, BGE-M3  $\rightarrow$  MiniLMv2-L6-H768, and Qwen3-Embedding-0.6B  $\rightarrow$  MiniLMv2-L6-H384 (Devlin et al., 2019; Wang et al., 2021; Chen et al., 2024; Zhang et al., 2025). These settings span students from 22M to 109M parameters and embedding bottlenecks of 2560  $\rightarrow$  768, 1024  $\rightarrow$  768, and 1024  $\rightarrow$  384, respectively. The Qwen teachers use last-token pooling, whereas BGE-M3 uses CLS pooling.

**Distillation corpus.** We construct the distillation corpus by sampling approximately 5K raw texts from each of the Emotion, WiC, and STS-B training sets. No labels or task annotations are used during distillation. We regard these three benchmarks as in-domain (IOD) and the remaining six evaluation benchmarks as out-of-domain (OOD). Every method uses the same corpus.

**Baselines.** We compare GATE-KD with methods covering complementary transfer mechanisms. RKD matches inter-example relations (Park et al., 2019), while Stella uses a multi-stage embedding-distillation recipe (Zhang et al., 2024a). CDM and DSKD address cross-tokenizer alignment and dual-space transfer, respectively (Chen et al., 2025; Zhang et al., 2024b). EMO combines token-level relations with hidden-state alignment (Truong et al., 2025), and TALAS combines teacher anchoring with hierarchy-based self-distillation (Dao et al., 2026). Method-specific training hyperparameters are reported in Appendix A.

**Evaluation protocol.** We evaluate frozen student embeddings on nine benchmarks from three task families. For classification on Banking77, Emotion, and Tweet, we fit a logistic-regression probe on each benchmark’s training split and report macro-F1. For pair classification on MRPC, SciTail, and WiC, we map cosine similarity to  $[0, 1]$  and report average precision. For semantic textual similarity on SICK, STS12, and STS-B, we report Spearman correlation with the gold scores. All metrics are multiplied by 100. Avg. is the unweighted mean of the nine primary scores, while Avg. IOD and Avg. OOD average the three in-domain and six out-of-domain benchmarks, respectively. We report means and sample standard deviations over three seeds.

### 4.2 MAIN RESULTS

Table 1 compares GATE-KD with six distillation baselines across nine benchmarks and three teacher–student configurations. GATE-KD achieves the highest overall average in all three configurations, improving over the next-best baseline by 0.38, 0.48, and 1.59 points in configurations (a), (b), and (c), respectively. The advantage is largest for the 22M-parameter student and therefore does not diminish in the most capacity-constrained configuration.

The task-level results show that this advantage is broad but not uniform. GATE-KD ranks first on 18 of the 27 individual benchmark comparisons, with its most consistent results on classification and semantic textual similarity. Other methods remain stronger on some pair-classification benchmarks, including MRPC and WiC. In the largest capacity-gap configuration, GATE-KD essentially matches TALAS in-domain, with 72.04 versus 72.19 Avg. IOD, while improving Avg. OOD from 79.85 to 80.49. The aggregate gain in this configuration is therefore driven by OOD performance rather than higher scores on the datasets supplying the distillation text.

Table 1: Main results across nine benchmarks. Distilled-student scores are mean  $\pm$  sample standard deviation over three seeds. Best and second-best results are shown in bold and underlined, respectively.

| Method                                                     | Classification (F1)     |                         |                         | Pair Classification (AP) |                         |                         | STS (Spearman)          |                         |                         | Averages                |                         |                         |
|------------------------------------------------------------|-------------------------|-------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|                                                            | Banking77               | Tweet                   | Emotion                 | MRPC                     | SciTail                 | WiC                     | SICK                    | STS12                   | STS-B                   | IOD                     | OOD                     | All                     |
| (a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)      |                         |                         |                         |                          |                         |                         |                         |                         |                         |                         |                         |                         |
| Teacher                                                    | 93.26                   | 71.04                   | 69.08                   | 84.80                    | 91.77                   | 66.79                   | 83.43                   | 82.55                   | 86.87                   | 74.25                   | 84.48                   | 81.07                   |
| Student base                                               | 84.33                   | 67.08                   | 47.79                   | 77.36                    | 67.74                   | 59.22                   | 42.43                   | 21.54                   | 20.29                   | 42.43                   | 60.08                   | 54.20                   |
| RKD                                                        | 89.76 $\pm$ 0.11        | 70.90 $\pm$ 0.31        | 59.98 $\pm$ 0.05        | 81.94 $\pm$ 0.26         | 75.59 $\pm$ 0.13        | 69.02 $\pm$ 0.18        | 72.93 $\pm$ 0.07        | 67.84 $\pm$ 0.18        | 73.72 $\pm$ 0.08        | 67.57 $\pm$ 0.05        | 76.49 $\pm$ 0.07        | 73.52 $\pm$ 0.06        |
| Stella                                                     | 88.48 $\pm$ 0.30        | 68.94 $\pm$ 0.17        | 53.69 $\pm$ 0.50        | 83.95 $\pm$ 0.10         | 75.39 $\pm$ 0.54        | <b>71.93</b> $\pm$ 0.09 | 69.87 $\pm$ 0.64        | 64.39 $\pm$ 0.21        | 70.86 $\pm$ 0.36        | 65.49 $\pm$ 0.17        | 75.17 $\pm$ 0.13        | 71.95 $\pm$ 0.06        |
| CDM                                                        | 88.13 $\pm$ 0.23        | 69.03 $\pm$ 0.20        | 52.83 $\pm$ 0.50        | 83.00 $\pm$ 0.06         | 77.10 $\pm$ 0.27        | 71.84 $\pm$ 0.09        | 67.89 $\pm$ 0.37        | 61.06 $\pm$ 0.37        | 69.19 $\pm$ 0.42        | 64.62 $\pm$ 0.28        | 74.37 $\pm$ 0.12        | 71.12 $\pm$ 0.18        |
| DSKD                                                       | 87.97 $\pm$ 0.16        | 68.90 $\pm$ 0.28        | 53.07 $\pm$ 0.31        | 83.03 $\pm$ 0.15         | 76.98 $\pm$ 0.59        | <u>71.86</u> $\pm$ 0.08 | 67.77 $\pm$ 0.66        | 61.11 $\pm$ 0.26        | 69.55 $\pm$ 0.12        | 64.83 $\pm$ 0.12        | 74.29 $\pm$ 0.14        | 71.14 $\pm$ 0.10        |
| EMO                                                        | 86.89 $\pm$ 0.07        | 68.99 $\pm$ 0.11        | 52.02 $\pm$ 1.09        | 79.74 $\pm$ 0.12         | 76.37 $\pm$ 0.35        | 69.62 $\pm$ 0.16        | 67.89 $\pm$ 0.09        | 56.46 $\pm$ 0.47        | 64.54 $\pm$ 0.50        | 62.06 $\pm$ 0.49        | 72.72 $\pm$ 0.15        | 69.17 $\pm$ 0.24        |
| TALAS                                                      | <u>90.53</u> $\pm$ 0.09 | <u>73.77</u> $\pm$ 0.21 | <u>68.47</u> $\pm$ 0.34 | <b>85.38</b> $\pm$ 0.05  | <b>81.83</b> $\pm$ 0.21 | 69.59 $\pm$ 0.14        | <u>76.79</u> $\pm$ 0.06 | <u>70.79</u> $\pm$ 0.19 | <b>78.50</b> $\pm$ 0.08 | <b>72.19</b> $\pm$ 0.07 | <u>79.85</u> $\pm$ 0.07 | <u>77.29</u> $\pm$ 0.04 |
| GATE-KD                                                    | <b>90.60</b> $\pm$ 0.21 | <b>73.86</b> $\pm$ 0.10 | <b>69.48</b> $\pm$ 0.22 | <u>85.31</u> $\pm$ 0.11  | <u>80.23</u> $\pm$ 0.13 | 68.19 $\pm$ 0.12        | <b>77.80</b> $\pm$ 0.14 | <b>75.15</b> $\pm$ 0.07 | <u>78.44</u> $\pm$ 0.16 | <u>72.04</u> $\pm$ 0.08 | <b>80.49</b> $\pm$ 0.03 | <b>77.67</b> $\pm$ 0.03 |
| (b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)               |                         |                         |                         |                          |                         |                         |                         |                         |                         |                         |                         |                         |
| Teacher                                                    | 93.49                   | 73.75                   | 69.03                   | 85.80                    | 91.87                   | 61.46                   | 79.18                   | 78.74                   | 84.86                   | 71.78                   | 83.81                   | 79.80                   |
| Student base                                               | 81.18                   | 71.30                   | 48.85                   | 79.66                    | 65.06                   | 60.89                   | 49.46                   | 26.76                   | 24.12                   | 44.62                   | 62.24                   | 56.36                   |
| RKD                                                        | 89.14 $\pm$ 0.09        | 73.29 $\pm$ 0.09        | <u>62.23</u> $\pm$ 0.27 | 84.35 $\pm$ 0.06         | 78.74 $\pm$ 0.11        | 65.02 $\pm$ 0.10        | 73.20 $\pm$ 0.19        | 65.02 $\pm$ 0.29        | 71.97 $\pm$ 0.38        | 66.41 $\pm$ 0.07        | 77.29 $\pm$ 0.06        | 73.66 $\pm$ 0.04        |
| Stella                                                     | 86.82 $\pm$ 0.28        | 70.06 $\pm$ 0.12        | 56.61 $\pm$ 0.74        | 83.02 $\pm$ 0.12         | 74.63 $\pm$ 0.12        | <b>69.72</b> $\pm$ 0.05 | 69.08 $\pm$ 0.20        | 63.14 $\pm$ 0.28        | 67.10 $\pm$ 0.10        | 64.47 $\pm$ 0.27        | 74.46 $\pm$ 0.05        | 71.13 $\pm$ 0.11        |
| CDM                                                        | 87.82 $\pm$ 0.19        | 69.99 $\pm$ 0.09        | 58.32 $\pm$ 0.51        | 83.44 $\pm$ 0.07         | 75.56 $\pm$ 0.40        | <u>69.70</u> $\pm$ 0.02 | 65.93 $\pm$ 0.16        | 60.84 $\pm$ 0.34        | 65.31 $\pm$ 0.15        | 64.45 $\pm$ 0.13        | 73.93 $\pm$ 0.06        | 70.77 $\pm$ 0.02        |
| DSKD                                                       | 87.70 $\pm$ 0.16        | 70.35 $\pm$ 0.32        | 59.71 $\pm$ 0.61        | 82.99 $\pm$ 0.18         | 75.58 $\pm$ 0.12        | 69.59 $\pm$ 0.04        | 65.36 $\pm$ 0.37        | 61.21 $\pm$ 0.31        | 65.02 $\pm$ 0.18        | 64.77 $\pm$ 0.27        | 73.87 $\pm$ 0.10        | 70.83 $\pm$ 0.11        |
| EMO                                                        | 87.54 $\pm$ 0.18        | 71.03 $\pm$ 0.14        | 58.52 $\pm$ 0.30        | 82.46 $\pm$ 0.18         | 74.73 $\pm$ 0.58        | 69.20 $\pm$ 0.22        | 64.71 $\pm$ 0.55        | 56.89 $\pm$ 0.27        | 61.25 $\pm$ 0.03        | 62.99 $\pm$ 0.19        | 72.89 $\pm$ 0.10        | 69.59 $\pm$ 0.12        |
| TALAS                                                      | <u>89.83</u> $\pm$ 0.09 | <b>74.70</b> $\pm$ 0.09 | 61.65 $\pm$ 0.23        | <b>86.62</b> $\pm$ 0.05  | <u>80.85</u> $\pm$ 0.22 | 66.72 $\pm$ 0.07        | <u>75.40</u> $\pm$ 0.10 | <u>67.08</u> $\pm$ 0.31 | <u>77.08</u> $\pm$ 0.01 | <u>68.48</u> $\pm$ 0.07 | <u>79.08</u> $\pm$ 0.08 | <u>75.55</u> $\pm$ 0.06 |
| GATE-KD                                                    | <b>90.17</b> $\pm$ 0.05 | <u>73.74</u> $\pm$ 0.24 | <b>64.83</b> $\pm$ 0.72 | <u>86.42</u> $\pm$ 0.09  | <b>80.97</b> $\pm$ 0.08 | 65.47 $\pm$ 0.09        | <b>76.77</b> $\pm$ 0.03 | <b>68.27</b> $\pm$ 0.04 | <b>77.63</b> $\pm$ 0.12 | <b>69.31</b> $\pm$ 0.21 | <b>79.39</b> $\pm$ 0.06 | <b>76.03</b> $\pm$ 0.11 |
| (c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M) |                         |                         |                         |                          |                         |                         |                         |                         |                         |                         |                         |                         |
| Teacher                                                    | 92.51                   | 71.28                   | 65.80                   | 83.38                    | 90.09                   | 66.84                   | 80.65                   | 77.11                   | 84.59                   | 72.41                   | 82.50                   | 79.14                   |
| Student base                                               | 67.03                   | 69.45                   | 42.90                   | 78.39                    | 64.55                   | 59.58                   | 47.60                   | 21.96                   | 22.08                   | 41.52                   | 58.16                   | 52.62                   |
| RKD                                                        | 86.76 $\pm$ 0.07        | 69.87 $\pm$ 0.21        | 56.31 $\pm$ 0.35        | 81.70 $\pm$ 0.03         | 76.03 $\pm$ 0.20        | 68.56 $\pm$ 0.22        | 69.21 $\pm$ 0.18        | 64.26 $\pm$ 0.19        | 70.25 $\pm$ 0.21        | 65.04 $\pm$ 0.20        | 74.64 $\pm$ 0.09        | 71.44 $\pm$ 0.10        |
| Stella                                                     | 84.04 $\pm$ 0.04        | 68.73 $\pm$ 0.44        | 52.41 $\pm$ 0.28        | 81.50 $\pm$ 0.12         | 72.26 $\pm$ 0.32        | <b>68.65</b> $\pm$ 0.16 | 64.64 $\pm$ 0.19        | 60.59 $\pm$ 0.12        | 64.27 $\pm$ 0.20        | 61.78 $\pm$ 0.13        | 71.96 $\pm$ 0.13        | 68.57 $\pm$ 0.10        |
| CDM                                                        | 84.74 $\pm$ 0.21        | 69.52 $\pm$ 0.20        | 53.76 $\pm$ 0.95        | 81.15 $\pm$ 0.14         | 72.86 $\pm$ 0.30        | 68.29 $\pm$ 0.11        | 63.98 $\pm$ 0.36        | 60.43 $\pm$ 0.22        | 63.52 $\pm$ 0.24        | 61.86 $\pm$ 0.43        | 72.11 $\pm$ 0.19        | 68.69 $\pm$ 0.27        |
| DSKD                                                       | 84.76 $\pm$ 0.10        | 69.56 $\pm$ 0.20        | 53.14 $\pm$ 0.29        | 81.15 $\pm$ 0.01         | 72.81 $\pm$ 0.09        | 68.33 $\pm$ 0.04        | 63.78 $\pm$ 0.06        | 60.24 $\pm$ 0.19        | 63.53 $\pm$ 0.18        | 61.67 $\pm$ 0.03        | 72.05 $\pm$ 0.06        | 68.59 $\pm$ 0.04        |
| EMO                                                        | 83.54 $\pm$ 0.31        | 69.31 $\pm$ 0.06        | 54.48 $\pm$ 0.27        | 80.87 $\pm$ 0.13         | 72.00 $\pm$ 0.16        | 67.89 $\pm$ 0.09        | 62.19 $\pm$ 0.16        | 57.04 $\pm$ 0.17        | 61.24 $\pm$ 0.23        | 61.20 $\pm$ 0.09        | 70.82 $\pm$ 0.07        | 67.62 $\pm$ 0.02        |
| TALAS                                                      | 86.39 $\pm$ 0.13        | <u>71.81</u> $\pm$ 0.11 | <u>58.40</u> $\pm$ 0.30 | <b>84.25</b> $\pm$ 0.03  | <u>79.29</u> $\pm$ 0.11 | 66.82 $\pm$ 0.02        | <u>70.42</u> $\pm$ 0.10 | <u>68.08</u> $\pm$ 0.22 | <u>73.34</u> $\pm$ 0.10 | <u>66.19</u> $\pm$ 0.11 | <u>76.71</u> $\pm$ 0.04 | <u>73.20</u> $\pm$ 0.06 |
| GATE-KD                                                    | <b>87.65</b> $\pm$ 0.06 | <b>72.32</b> $\pm$ 0.24 | <b>63.04</b> $\pm$ 0.28 | <u>84.20</u> $\pm$ 0.06  | <b>81.60</b> $\pm$ 0.05 | 68.07 $\pm$ 0.12        | <b>72.48</b> $\pm$ 0.08 | <b>68.96</b> $\pm$ 0.18 | <b>74.76</b> $\pm$ 0.14 | <b>68.62</b> $\pm$ 0.21 | <b>77.87</b> $\pm$ 0.02 | <b>74.79</b> $\pm$ 0.06 |

### 4.3 MEASURED OPTIMIZATION COST

Table 2 reports the cost of student optimization under the same execution protocol on NVIDIA H200 GPUs. Step time and throughput exclude the first ten warm-up steps. GPU optimization time sums CUDA-event durations for the minibatch forward, loss, backward, and optimizer update. It therefore includes the topology computation performed inside GATE-KD’s training steps, but excludes model and tokenizer loading, data preprocessing, final evaluation, teacher-cache construction, the one-time PCA fit, and the four student re-encoding passes used for gauge refitting. The table measures the cost of the optimization loop rather than end-to-end wall-clock cost.

Teacher access also differs across method families. GATE-KD, RKD, Stella, and TALAS read normalized final teacher embeddings from the same reusable cache and do not execute the teacher during student optimization. CDM, DSKD, and EMO execute the teacher online; EMO additionally consumes teacher hidden states and attention. Charging a shared cache entirely to the first cached method would be arbitrary, while excluding it understates the cost of a standalone run. We therefore report the cache-independent optimization measurement and do not interpret it as total training cost.

GATE-KD uses 26%, 31%, and 38% of TALAS’s measured GPU optimization time in configurations (a), (b), and (c), respectively, while also attaining higher overall scores. The improvement is primarily in throughput rather than peak memory allocation, for which TALAS remains lower across the three configurations. A controlled end-to-end comparison would additionally require isolated runs that separately time teacher encoding, cache I/O, PCA and gauge updates, preprocessing, and evaluation; the present measurements do not support such a claim.

Table 2: Measured student-optimization cost on NVIDIA H200 GPUs. Times exclude teacher-cache construction, setup, gauge-refit passes, and evaluation; they are not end-to-end wall-clock measurements.

| Method                                                                       | Avg. $\uparrow$         | ms/step $\downarrow$  | samples/s $\uparrow$      | GPU opt. min $\downarrow$ | peak GiB $\downarrow$ |
|------------------------------------------------------------------------------|-------------------------|-----------------------|---------------------------|---------------------------|-----------------------|
| <i>(a) Qwen3-Embedding-4B <math>\rightarrow</math> BERT-base (109M)</i>      |                         |                       |                           |                           |                       |
| RKD                                                                          | 73.52 $\pm$ 0.06        | 119.7 $\pm$ 18.3      | 1081.0 $\pm$ 179.8        | 1.20 $\pm$ 0.18           | 1.72 $\pm$ 0.00       |
| Stella                                                                       | 71.95 $\pm$ 0.06        | 130.7 $\pm$ 8.1       | 975.9 $\pm$ 58.8          | 1.27 $\pm$ 0.08           | 1.78 $\pm$ 0.01       |
| CDM                                                                          | 71.12 $\pm$ 0.18        | 340.4 $\pm$ 13.6      | 374.2 $\pm$ 15.3          | 3.38 $\pm$ 0.12           | 9.31 $\pm$ 0.01       |
| DSKD                                                                         | 71.14 $\pm$ 0.10        | 290.9 $\pm$ 9.0       | 437.7 $\pm$ 13.3          | 2.88 $\pm$ 0.09           | 9.45 $\pm$ 0.00       |
| EMO                                                                          | 69.17 $\pm$ 0.24        | 721.1 $\pm$ 12.9      | 176.5 $\pm$ 3.1           | 7.02 $\pm$ 0.12           | 9.30 $\pm$ 0.00       |
| TALAS                                                                        | 77.29 $\pm$ 0.04        | 329.0 $\pm$ 12.5      | 387.1 $\pm$ 15.0          | 3.23 $\pm$ 0.11           | 1.43 $\pm$ 0.00       |
| <b>GATE-KD</b>                                                               | <b>77.67</b> $\pm$ 0.03 | <b>83.4</b> $\pm$ 3.8 | <b>1527.2</b> $\pm$ 70.8  | <b>0.84</b> $\pm$ 0.04    | 1.72 $\pm$ 0.00       |
| <i>(b) BGE-M3 <math>\rightarrow</math> MiniLMv2-H768 (66M)</i>               |                         |                       |                           |                           |                       |
| RKD                                                                          | 73.66 $\pm$ 0.04        | 77.5 $\pm$ 8.8        | 1656.4 $\pm$ 201.2        | 0.79 $\pm$ 0.09           | 1.07 $\pm$ 0.00       |
| Stella                                                                       | 71.13 $\pm$ 0.11        | 152.8 $\pm$ 30.8      | 859.1 $\pm$ 195.6         | 1.49 $\pm$ 0.30           | 1.09 $\pm$ 0.00       |
| CDM                                                                          | 70.77 $\pm$ 0.02        | 127.5 $\pm$ 13.8      | 1006.4 $\pm$ 115.8        | 1.30 $\pm$ 0.12           | 2.15 $\pm$ 0.01       |
| DSKD                                                                         | 70.83 $\pm$ 0.11        | 115.0 $\pm$ 21.9      | 1137.2 $\pm$ 241.0        | 1.14 $\pm$ 0.20           | 2.22 $\pm$ 0.01       |
| EMO                                                                          | 69.59 $\pm$ 0.12        | 394.6 $\pm$ 33.4      | 324.0 $\pm$ 28.3          | 3.87 $\pm$ 0.35           | 2.15 $\pm$ 0.01       |
| TALAS                                                                        | 75.55 $\pm$ 0.06        | 184.4 $\pm$ 4.6       | 690.2 $\pm$ 17.3          | 1.82 $\pm$ 0.03           | 0.85 $\pm$ 0.00       |
| <b>GATE-KD</b>                                                               | <b>76.03</b> $\pm$ 0.11 | <b>56.5</b> $\pm$ 7.7 | <b>2281.6</b> $\pm$ 310.0 | <b>0.57</b> $\pm$ 0.07    | 1.07 $\pm$ 0.00       |
| <i>(c) Qwen3-Embedding-0.6B <math>\rightarrow</math> MiniLMv2-H384 (22M)</i> |                         |                       |                           |                           |                       |
| RKD                                                                          | 71.44 $\pm$ 0.10        | 64.8 $\pm$ 13.7       | 2012.3 $\pm$ 488.6        | 0.68 $\pm$ 0.14           | 0.41 $\pm$ 0.00       |
| Stella                                                                       | 68.57 $\pm$ 0.10        | 150.7 $\pm$ 15.3      | 850.0 $\pm$ 81.5          | 1.47 $\pm$ 0.13           | 0.42 $\pm$ 0.00       |
| CDM                                                                          | 68.69 $\pm$ 0.27        | 161.8 $\pm$ 21.6      | 796.3 $\pm$ 113.1         | 1.65 $\pm$ 0.20           | 1.53 $\pm$ 0.00       |
| DSKD                                                                         | 68.59 $\pm$ 0.04        | 175.4 $\pm$ 8.2       | 726.4 $\pm$ 33.5          | 1.78 $\pm$ 0.06           | 1.56 $\pm$ 0.00       |
| EMO                                                                          | 67.62 $\pm$ 0.02        | 499.3 $\pm$ 1.4       | 254.8 $\pm$ 0.7           | 4.90 $\pm$ 0.01           | 1.53 $\pm$ 0.00       |
| TALAS                                                                        | 73.20 $\pm$ 0.06        | 169.8 $\pm$ 15.4      | 753.7 $\pm$ 71.4          | 1.69 $\pm$ 0.14           | 0.34 $\pm$ 0.00       |
| <b>GATE-KD</b>                                                               | <b>74.79</b> $\pm$ 0.06 | <b>62.9</b> $\pm$ 3.1 | <b>2025.2</b> $\pm$ 98.3  | <b>0.64</b> $\pm$ 0.04    | 0.41 $\pm$ 0.00       |

## 5 ANALYSIS

### 5.1 COORDINATE COMPARABILITY AT THE BOTTLENECK

Table 3 isolates the endpoint interface in configuration (c) by holding the endpoint-only objective and training protocol fixed. Within the same PCA subspace, the unaligned coordinates reach 69.35 Avg., a random orthogonal orientation reaches 70.37, and the fitted Procrustes gauge reaches 74.28. Procrustes alignment therefore improves the unaligned PCA interface by 4.93 points without changing either the retained subspace or its pairwise geometry. The fitted gauge also exceeds the random orientation by 3.91 points, indicating that its benefit comes from alignment with the student rather than from rotating the target coordinates alone.

The fixed PCA–Procrustes interface also outperforms learned maps in both directions, which reach 61.46 for teacher-to-student mapping and 69.33 for student-to-teacher mapping. This result is consistent with avoiding a mapping that is updated jointly with the student by gradient descent. Retained energy alone does not explain downstream performance. The random subspace retains 0.381 of the teacher energy, compared with 0.928 for PCA, yet exceeds unaligned PCA by 1.57 points. Together, these controls show that a high-energy subspace is not an effective pointwise target unless its coordinates are made comparable to the student. The random controls use a single map draw, so they do not establish a general ranking between PCA and randomly selected subspaces.



Table 3: Endpoint-interface ablation in configuration (c). All variants optimize  $\mathcal{L}_{\text{end}}$  only. Scores are mean  $\pm$  sample standard deviation over three seeds. Random controls use one map draw shared across the training seeds.

| Target map                | Gauge             | Retained energy $\uparrow$ | Avg. IOD $\uparrow$                | Avg. OOD $\uparrow$                | Avg. $\uparrow$                    |
|---------------------------|-------------------|----------------------------|------------------------------------|------------------------------------|------------------------------------|
| Learned $T \rightarrow S$ | —                 | N/A                        | $54.34 \pm 0.68$                   | $65.02 \pm 0.25$                   | $61.46 \pm 0.39$                   |
| Learned $S \rightarrow T$ | —                 | N/A                        | $62.54 \pm 0.36$                   | $72.72 \pm 0.31$                   | $69.33 \pm 0.32$                   |
| PCA                       | None              | 0.928                      | $62.49 \pm 0.19$                   | $72.78 \pm 0.07$                   | $69.35 \pm 0.04$                   |
| PCA                       | Random orthogonal | 0.928                      | $63.68 \pm 0.10$                   | $73.72 \pm 0.09$                   | $70.37 \pm 0.08$                   |
| Random orthogonal         | None              | 0.381                      | $64.44 \pm 0.14$                   | $74.16 \pm 0.04$                   | $70.92 \pm 0.07$                   |
| <b>PCA</b>                | <b>Procrustes</b> | <b>0.928</b>               | <b><math>67.76 \pm 0.05</math></b> | <b><math>77.54 \pm 0.07</math></b> | <b><math>74.28 \pm 0.03</math></b> |

## 5.2 COMPLEMENTARY ENDPOINT AND TOPOLOGY SIGNALS

Table 4 separates coordinate-dependent endpoint supervision from coordinate-invariant topology transfer. The endpoint-only objective provides most of the downstream improvement, raising Avg. from 52.62 for the base student to 74.28. In contrast,  $H_0$ -only supervision reaches 64.22 while reducing the topology residual from 0.3595 to 0.1240. The low residual but substantially lower downstream score shows that matching connectivity alone does not determine semantic correspondence. Combining the two signals improves the endpoint-only result by 0.51 points to 74.79 and lowers its topology residual from 0.2122 to 0.1281. Endpoint correspondence and topological structure therefore provide complementary supervision.

The remaining comparisons evaluate two refinements within the combined objective. Computing  $H_0$  from the original teacher space improves Avg. from 74.57 to 74.79 relative to projected teacher targets and produces a lower topology residual, 0.1281 versus 0.1355. Refitting the gauge after each epoch improves Avg. from 74.48 to 74.79 over retaining the alignment fitted at initialization. Both gains are concentrated more strongly in IOD performance and are modest relative to the contribution of the endpoint interface. These results support native-space topology and epoch-wise refitting as refinements of the complete objective rather than its primary source of improvement.

Table 4: Decomposition of endpoint supervision, topology transfer, and gauge refitting in configuration (c). Scores are mean  $\pm$  sample standard deviation over three seeds. The  $H_0$  residual is measured in the original teacher space.

| Configuration                    | $H_0$ source | Gauge refit | $H_0$ residual $\downarrow$           | Avg. IOD $\uparrow$                | Avg. OOD $\uparrow$                | Avg. $\uparrow$                    |
|----------------------------------|--------------|-------------|---------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Student base                     | —            | —           | $0.3595 \pm 0.0000$                   | $41.52 \pm 0.00$                   | $58.16 \pm 0.00$                   | $52.62 \pm 0.00$                   |
| $H_0$ only                       | Original     | —           | <b><math>0.1240 \pm 0.0012</math></b> | $58.00 \pm 1.01$                   | $67.33 \pm 0.62$                   | $64.22 \pm 0.73$                   |
| Endpoint only                    | —            | On          | $0.2122 \pm 0.0004$                   | $67.76 \pm 0.05$                   | $77.54 \pm 0.07$                   | $74.28 \pm 0.03$                   |
| Endpoint + $H_0$                 | Projected    | On          | $0.1355 \pm 0.0003$                   | $68.27 \pm 0.12$                   | $77.72 \pm 0.00$                   | $74.57 \pm 0.04$                   |
| Endpoint + $H_0$                 | Original     | Off         | $0.1309 \pm 0.0001$                   | $68.03 \pm 0.10$                   | $77.71 \pm 0.03$                   | $74.48 \pm 0.01$                   |
| Endpoint + $H_0$ ( <b>Ours</b> ) | Original     | On          | $0.1281 \pm 0.0003$                   | <b><math>68.62 \pm 0.21</math></b> | <b><math>77.87 \pm 0.02</math></b> | <b><math>74.79 \pm 0.06</math></b> |

## 5.3 SENSITIVITY TO TRAINING CHOICES

Figure 1 examines sensitivity to the topology weight, training batch size, and gauge-calibration size in configuration (c). Performance remains stable across moderate topology weights, with  $\lambda_{\text{topo}} = 0.5$  attaining the highest average and settings from 0.5 to 1.0 differing by at most 0.24 points. Reducing the gauge-calibration set from the full corpus to 2,048 examples changes Avg. by only 0.28 points, indicating that the alignment can be estimated from a relatively small sample. Smaller batches obtain higher scores, but this sweep fixes the number of epochs rather than optimizer updates. The batch comparison is therefore not compute matched and does not establish that smaller batches are intrinsically better. Complete numerical results are reported in Appendix B.

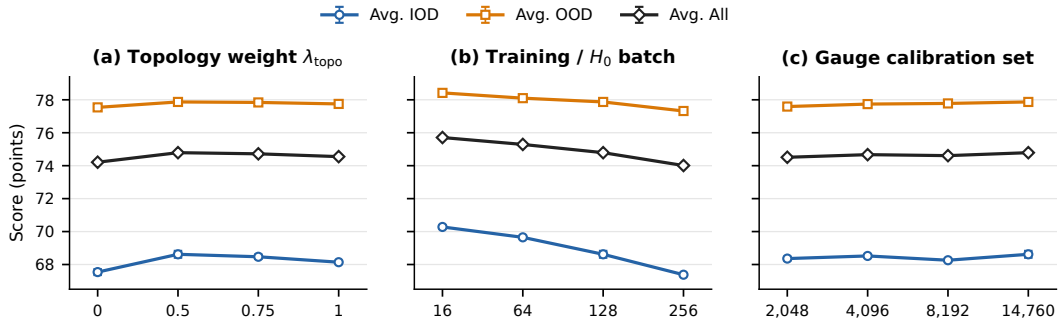


Figure 1: Sensitivity to topology weight, training batch size, and gauge-calibration size in configuration (c). Points and error bars show mean  $\pm$  sample standard deviation over three seeds. The batch sweep fixes the number of epochs and is not compute matched.

## 6 CONCLUSION

Cross-dimensional embedding distillation is not only a compression problem but also requires a well-defined interface between teacher and student coordinates. GATE-KD addresses this issue by combining a fixed PCA subspace with closed-form gauge alignment and native-space  $H_0$  matching, using only final embeddings. Across three teacher-student configurations, this terminal-only formulation achieves the highest aggregate scores among the evaluated distilled students while substantially reducing measured optimization time relative to a strong hierarchy-based baseline. Controlled analyses show that coordinate alignment matters beyond retained variance and that topology complements pointwise endpoint supervision without requiring a shared ambient space. These results support treating coordinate-dependent correspondence and coordinate-invariant structure as separate design problems in embedding distillation. The current sentence-level objective does not fully preserve the teacher’s zero-shot retrieval capability, leaving broader retrieval transfer for future work.

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## A TRAINING DETAILS

All runs use a maximum sequence length of 256 and seeds 42, 43, and 44. For GATE-KD,  $\lambda_{\text{topo}} = 0.5$ , the  $H_0$  cloud is the current training batch, and the gauge is refitted after each non-final epoch. Gauge calibration uses the full distillation corpus. Table 5 lists the method-specific training hyperparameters.

Table 5: Training hyperparameters for the main experiments.

| Method  | Epochs | Batch size | Learning rate      |
|---------|--------|------------|--------------------|
| RKD     | 5      | 128        | $7 \times 10^{-5}$ |
| Stella  | 2 + 3  | 128        | $5 \times 10^{-5}$ |
| CDM     | 5      | 128        | $2 \times 10^{-5}$ |
| DSKD    | 5      | 128        | $2 \times 10^{-5}$ |
| EMO     | 5      | 128        | $1 \times 10^{-5}$ |
| TALAS   | 5      | 128        | $2 \times 10^{-5}$ |
| GATE-KD | 5      | 128        | $7 \times 10^{-5}$ |

## B DETAILED SENSITIVITY RESULTS

Table 6 reports the numerical results underlying Figure 1. Settings not varied within each block remain at the main recipe. The  $H_0$  cloud is always the current training batch. The batch sweep fixes the number of epochs and is therefore not compute matched.

Table 6: Numerical sensitivity results for configuration (c).

| Factor                  | Value  | Avg. IOD $\uparrow$     | Avg. OOD $\uparrow$     | Avg. $\uparrow$         |
|-------------------------|--------|-------------------------|-------------------------|-------------------------|
| $\lambda_{\text{topo}}$ | 0      | 67.54 $\pm$ 0.18        | 77.54 $\pm$ 0.07        | 74.21 $\pm$ 0.03        |
|                         | 0.5    | <b>68.62</b> $\pm$ 0.21 | <b>77.87</b> $\pm$ 0.02 | <b>74.79</b> $\pm$ 0.06 |
|                         | 0.75   | 68.47 $\pm$ 0.13        | 77.84 $\pm$ 0.04        | 74.72 $\pm$ 0.04        |
|                         | 1      | 68.14 $\pm$ 0.06        | 77.75 $\pm$ 0.03        | 74.55 $\pm$ 0.03        |
| Training/ $H_0$ batch   | 16     | <b>70.28</b> $\pm$ 0.09 | <b>78.42</b> $\pm$ 0.09 | <b>75.71</b> $\pm$ 0.03 |
|                         | 64     | 69.65 $\pm$ 0.09        | 78.10 $\pm$ 0.08        | 75.29 $\pm$ 0.05        |
|                         | 128    | 68.62 $\pm$ 0.21        | 77.87 $\pm$ 0.02        | 74.79 $\pm$ 0.06        |
|                         | 256    | 67.38 $\pm$ 0.03        | 77.32 $\pm$ 0.07        | 74.01 $\pm$ 0.05        |
| Gauge calibration       | 2,048  | 68.36 $\pm$ 0.11        | 77.59 $\pm$ 0.04        | 74.51 $\pm$ 0.06        |
|                         | 4,096  | 68.52 $\pm$ 0.08        | 77.74 $\pm$ 0.05        | 74.67 $\pm$ 0.02        |
|                         | 8,192  | 68.26 $\pm$ 0.08        | 77.78 $\pm$ 0.03        | 74.61 $\pm$ 0.02        |
|                         | 14,760 | <b>68.62</b> $\pm$ 0.21 | <b>77.87</b> $\pm$ 0.02 | <b>74.79</b> $\pm$ 0.06 |

## C RESULTS AT A LARGER DISTILLATION SCALE

We repeat the complete method sweep on approximately 100K unlabeled texts drawn from the training splits of all nine sentence-level benchmarks. No labels or task annotations are used. Relative to the main comparison, this setting changes both corpus size and dataset coverage, so it evaluates robustness to broader distillation data rather than isolating data scale alone.

Table 7: Results at a larger distillation scale.

| Method                                         | Classification (F1) |            |            | Pair Classification (AP) |            |            | STS (Spearman) |            |            | Avg.       |
|------------------------------------------------|---------------------|------------|------------|--------------------------|------------|------------|----------------|------------|------------|------------|
|                                                | Banking77           | Tweet      | Emotion    | MRPC                     | SciTail    | WiC        | SICK           | STS12      | STS-B      |            |
| (a) Qwen3-Embedding-4B → BERT-base (109M)      |                     |            |            |                          |            |            |                |            |            |            |
| Teacher                                        | 93.26               | 71.04      | 69.08      | 84.80                    | 91.77      | 66.79      | 83.43          | 82.55      | 86.87      | 81.07      |
| Student base                                   | 84.33               | 67.08      | 47.79      | 77.36                    | 67.74      | 59.22      | 42.43          | 21.54      | 20.29      | 54.20      |
| RKD                                            | 92.23±0.15          | 71.96±0.18 | 63.79±0.40 | 85.00±0.04               | 80.27±0.38 | 66.31±0.10 | 79.32±0.02     | 71.72±0.16 | 77.70±0.20 | 76.48±0.04 |
| Stella                                         | 90.02±0.15          | 71.90±0.22 | 60.14±0.45 | 85.96±0.05               | 78.82±0.27 | 69.93±0.13 | 76.53±0.23     | 69.79±0.12 | 77.22±0.16 | 75.59±0.10 |
| CDM                                            | 89.16±0.04          | 69.63±0.23 | 53.74±0.13 | 85.57±0.08               | 71.22±0.19 | 70.04±0.05 | 71.26±0.29     | 66.97±0.21 | 73.69±0.38 | 72.36±0.13 |
| DSKD                                           | 88.87±0.07          | 69.05±0.09 | 54.02±0.56 | 85.90±0.05               | 70.02±0.31 | 69.63±0.21 | 71.07±0.10     | 66.52±0.17 | 73.52±0.01 | 72.07±0.03 |
| EMO                                            | 84.96±0.84          | 68.93±0.65 | 54.80±1.52 | 85.05±0.04               | 65.75±0.22 | 68.80±0.13 | 62.00±2.71     | 61.75±1.85 | 69.54±0.37 | 69.07±0.89 |
| TALAS                                          | 93.12±0.13          | 74.67±0.38 | 71.42±0.26 | 86.30±0.01               | 87.06±0.12 | 68.09±0.02 | 80.29±0.03     | 76.34±0.01 | 82.26±0.05 | 79.95±0.08 |
| GATE-KD                                        | 93.47±0.12          | 74.18±0.25 | 72.50±0.47 | 85.45±0.08               | 87.24±0.12 | 66.57±0.08 | 81.85±0.06     | 76.40±0.07 | 82.52±0.08 | 80.02±0.03 |
| (b) BGE-M3 → MiniLMv2-H768 (66M)               |                     |            |            |                          |            |            |                |            |            |            |
| Teacher                                        | 93.49               | 73.75      | 69.03      | 85.80                    | 91.87      | 61.46      | 79.18          | 78.74      | 84.86      | 79.80      |
| Student base                                   | 81.18               | 71.30      | 48.85      | 79.66                    | 65.06      | 60.89      | 49.46          | 26.76      | 24.12      | 56.36      |
| RKD                                            | 91.68±0.06          | 74.57±0.40 | 62.45±0.75 | 85.81±0.08               | 79.81±0.17 | 63.57±0.13 | 76.91±0.05     | 69.54±0.11 | 75.56±0.09 | 75.54±0.10 |
| Stella                                         | 88.72±0.41          | 73.43±0.23 | 61.63±0.29 | 86.39±0.08               | 74.13±0.31 | 66.98±0.27 | 73.32±0.07     | 68.30±0.12 | 72.66±0.14 | 73.95±0.04 |
| CDM                                            | 87.64±0.23          | 70.82±0.14 | 56.37±0.57 | 85.67±0.08               | 67.69±0.23 | 66.89±0.20 | 70.15±0.48     | 66.68±0.26 | 69.49±0.15 | 71.27±0.10 |
| DSKD                                           | 86.69±0.38          | 69.76±0.22 | 56.98±0.39 | 84.92±0.07               | 66.75±0.17 | 67.23±0.10 | 68.26±0.45     | 66.28±0.02 | 68.31±0.35 | 70.58±0.10 |
| EMO                                            | 87.17±0.59          | 71.07±0.22 | 59.32±0.02 | 84.25±0.41               | 67.89±0.92 | 66.09±0.18 | 69.04±0.38     | 62.26±1.77 | 66.47±0.40 | 70.40±0.35 |
| TALAS                                          | 92.44±0.10          | 75.58±0.15 | 65.13±0.21 | 87.09±0.09               | 85.39±0.14 | 64.39±0.12 | 78.14±0.06     | 72.56±0.13 | 81.03±0.05 | 77.97±0.03 |
| GATE-KD                                        | 92.56±0.04          | 76.30±0.31 | 65.62±0.29 | 86.58±0.05               | 87.20±0.06 | 63.43±0.02 | 79.23±0.03     | 72.69±0.05 | 80.60±0.13 | 78.24±0.05 |
| (c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M) |                     |            |            |                          |            |            |                |            |            |            |
| Teacher                                        | 92.51               | 71.28      | 65.80      | 83.38                    | 90.09      | 66.84      | 80.65          | 77.11      | 84.59      | 79.14      |
| Student base                                   | 67.03               | 69.45      | 42.90      | 78.39                    | 64.55      | 59.58      | 47.60          | 21.96      | 22.08      | 52.62      |
| RKD                                            | 89.74±0.04          | 72.34±0.22 | 57.99±0.52 | 84.13±0.12               | 75.20±0.14 | 67.49±0.06 | 74.48±0.03     | 68.12±0.15 | 72.55±0.14 | 73.56±0.09 |
| Stella                                         | 85.74±0.22          | 70.25±0.07 | 53.48±0.28 | 84.43±0.04               | 71.23±0.05 | 66.51±0.14 | 70.96±0.28     | 63.02±0.04 | 67.42±0.24 | 70.34±0.02 |
| CDM                                            | 85.18±0.08          | 69.14±0.07 | 52.30±0.26 | 84.02±0.04               | 68.24±0.23 | 66.57±0.29 | 69.15±0.13     | 63.05±0.06 | 66.55±0.10 | 69.36±0.08 |
| DSKD                                           | 85.04±0.14          | 69.21±0.13 | 52.85±0.36 | 84.34±0.10               | 67.33±0.24 | 65.98±0.12 | 69.01±0.13     | 63.46±0.21 | 66.42±0.09 | 69.29±0.09 |
| EMO                                            | 85.08±0.17          | 68.11±0.10 | 53.96±0.60 | 84.38±0.16               | 65.48±0.16 | 65.23±0.17 | 68.61±0.07     | 60.00±0.82 | 65.38±0.38 | 68.47±0.15 |
| TALAS                                          | 89.79±0.03          | 73.87±0.17 | 61.75±0.61 | 85.99±0.03               | 82.07±0.09 | 65.79±0.04 | 75.67±0.05     | 73.91±0.03 | 78.31±0.04 | 76.35±0.08 |
| GATE-KD                                        | 91.68±0.09          | 74.43±0.21 | 66.88±0.36 | 85.00±0.06               | 83.46±0.02 | 67.44±0.04 | 77.94±0.06     | 72.17±0.02 | 78.36±0.07 | 77.48±0.04 |

Table 7 shows that GATE-KD attains the highest mean Avg. in all three teacher–student configurations. It exceeds TALAS by 0.07, 0.27, and 1.13 points in configurations (a), (b), and (c), respectively. The margin is narrow for the two larger students and remains largest for the compact student.

## D ZERO-SHOT RETRIEVAL

We evaluate the models distilled in the main comparison on ArguAna, SciFact, SciDocs, NFCorpus, and FiQA-2018 using exhaustive cosine ranking. Evaluation uses no instruction prefix or task-specific retrieval training and reports nDCG@10. Retrieval scores are not included in the sentence-level averages of Table 1. Teacher and student-base scores are deterministic, while distilled-student scores are mean  $\pm$  sample standard deviation over three seeds.

Table 8: Zero-shot retrieval on five BEIR benchmarks after distillation, measured by nDCG@10.

| Method                                                                       | ArguAna                 | SciFact                 | SciDocs                | NFCorpus               | FiQA-2018              | Avg.                    |
|------------------------------------------------------------------------------|-------------------------|-------------------------|------------------------|------------------------|------------------------|-------------------------|
| <i>(a) Qwen3-Embedding-4B <math>\rightarrow</math> BERT-base (109M)</i>      |                         |                         |                        |                        |                        |                         |
| Teacher                                                                      | 74.66                   | 73.75                   | 25.69                  | 36.23                  | 50.15                  | 52.10                   |
| Student base                                                                 | 12.65                   | 0.89                    | 0.38                   | 1.33                   | 0.05                   | 3.06                    |
| RKD                                                                          | 39.68 $\pm$ 0.25        | 9.74 $\pm$ 0.64         | 1.19 $\pm$ 0.04        | 4.30 $\pm$ 0.30        | 2.82 $\pm$ 0.09        | 11.55 $\pm$ 0.18        |
| Stella                                                                       | 39.12 $\pm$ 0.17        | 14.29 $\pm$ 0.17        | 3.79 $\pm$ 0.26        | 6.21 $\pm$ 0.14        | 5.57 $\pm$ 0.04        | 13.79 $\pm$ 0.03        |
| CDM                                                                          | 33.62 $\pm$ 0.27        | 15.11 $\pm$ 0.20        | 3.28 $\pm$ 0.14        | 6.26 $\pm$ 0.09        | 5.08 $\pm$ 0.22        | 12.67 $\pm$ 0.04        |
| DSKD                                                                         | 33.70 $\pm$ 0.39        | 15.75 $\pm$ 0.49        | 3.35 $\pm$ 0.11        | 6.21 $\pm$ 0.17        | 5.34 $\pm$ 0.24        | 12.87 $\pm$ 0.27        |
| EMO                                                                          | 21.90 $\pm$ 0.34        | 8.84 $\pm$ 0.28         | 1.87 $\pm$ 0.11        | 4.87 $\pm$ 0.16        | 2.04 $\pm$ 0.06        | 7.91 $\pm$ 0.07         |
| TALAS                                                                        | <b>48.95</b> $\pm$ 0.17 | <b>19.94</b> $\pm$ 0.09 | <b>5.70</b> $\pm$ 0.12 | <b>7.94</b> $\pm$ 0.08 | <b>8.15</b> $\pm$ 0.02 | <b>18.14</b> $\pm$ 0.07 |
| <b>GATE-KD</b>                                                               | <b>45.20</b> $\pm$ 0.32 | <b>19.25</b> $\pm$ 0.66 | 2.81 $\pm$ 0.09        | 6.79 $\pm$ 0.22        | 6.56 $\pm$ 0.21        | 16.12 $\pm$ 0.26        |
| <i>(b) BGE-M3 <math>\rightarrow</math> MiniLMv2-H768 (66M)</i>               |                         |                         |                        |                        |                        |                         |
| Teacher                                                                      | 54.01                   | 64.16                   | 16.37                  | 31.80                  | 41.12                  | 41.49                   |
| Student base                                                                 | 9.32                    | 1.15                    | 0.44                   | 1.22                   | 0.08                   | 2.44                    |
| RKD                                                                          | 21.27 $\pm$ 0.23        | 7.84 $\pm$ 0.12         | 1.96 $\pm$ 0.05        | 4.31 $\pm$ 0.14        | 2.26 $\pm$ 0.03        | 7.53 $\pm$ 0.09         |
| Stella                                                                       | 18.46 $\pm$ 0.37        | 3.11 $\pm$ 0.27         | 1.25 $\pm$ 0.06        | 2.29 $\pm$ 0.15        | 1.66 $\pm$ 0.08        | 5.35 $\pm$ 0.16         |
| CDM                                                                          | 15.83 $\pm$ 0.19        | 3.77 $\pm$ 0.38         | 0.88 $\pm$ 0.02        | 2.88 $\pm$ 0.16        | 1.41 $\pm$ 0.21        | 4.95 $\pm$ 0.15         |
| DSKD                                                                         | 16.25 $\pm$ 0.15        | 4.05 $\pm$ 0.10         | 0.88 $\pm$ 0.03        | 2.96 $\pm$ 0.11        | 1.59 $\pm$ 0.07        | 5.15 $\pm$ 0.07         |
| EMO                                                                          | 13.10 $\pm$ 0.27        | 2.83 $\pm$ 0.17         | 0.72 $\pm$ 0.04        | 2.25 $\pm$ 0.07        | 0.85 $\pm$ 0.09        | 3.95 $\pm$ 0.08         |
| TALAS                                                                        | <b>30.14</b> $\pm$ 0.32 | 6.55 $\pm$ 0.34         | 1.90 $\pm$ 0.08        | 2.82 $\pm$ 0.18        | 2.83 $\pm$ 0.09        | 8.85 $\pm$ 0.17         |
| <b>GATE-KD</b>                                                               | <b>25.30</b> $\pm$ 0.02 | <b>11.22</b> $\pm$ 0.27 | <b>3.31</b> $\pm$ 0.02 | <b>4.93</b> $\pm$ 0.06 | <b>4.83</b> $\pm$ 0.01 | <b>9.92</b> $\pm$ 0.05  |
| <i>(c) Qwen3-Embedding-0.6B <math>\rightarrow</math> MiniLMv2-H384 (22M)</i> |                         |                         |                        |                        |                        |                         |
| Teacher                                                                      | 68.11                   | 68.26                   | 15.72                  | 29.73                  | 38.68                  | 44.10                   |
| Student base                                                                 | 9.85                    | 0.66                    | 0.32                   | 1.28                   | 0.05                   | 2.43                    |
| RKD                                                                          | 30.12 $\pm$ 0.79        | 5.57 $\pm$ 0.76         | 1.77 $\pm$ 0.12        | 3.34 $\pm$ 0.07        | 1.88 $\pm$ 0.06        | 8.54 $\pm$ 0.33         |
| Stella                                                                       | 24.49 $\pm$ 0.45        | 2.65 $\pm$ 0.09         | 1.09 $\pm$ 0.07        | 2.62 $\pm$ 0.07        | 1.01 $\pm$ 0.11        | 6.37 $\pm$ 0.09         |
| CDM                                                                          | 21.09 $\pm$ 0.30        | 1.66 $\pm$ 0.17         | 1.13 $\pm$ 0.03        | 2.57 $\pm$ 0.05        | 0.89 $\pm$ 0.02        | 5.47 $\pm$ 0.06         |
| DSKD                                                                         | 21.56 $\pm$ 0.38        | 1.77 $\pm$ 0.20         | 1.18 $\pm$ 0.06        | 2.63 $\pm$ 0.03        | 0.89 $\pm$ 0.07        | 5.61 $\pm$ 0.13         |
| EMO                                                                          | 17.18 $\pm$ 0.14        | 1.41 $\pm$ 0.09         | 0.92 $\pm$ 0.02        | 1.96 $\pm$ 0.06        | 0.70 $\pm$ 0.03        | 4.44 $\pm$ 0.05         |
| TALAS                                                                        | 35.21 $\pm$ 0.52        | 4.17 $\pm$ 0.09         | 1.44 $\pm$ 0.04        | 3.12 $\pm$ 0.10        | 1.87 $\pm$ 0.07        | 9.16 $\pm$ 0.14         |
| <b>GATE-KD</b>                                                               | <b>36.18</b> $\pm$ 0.09 | <b>10.26</b> $\pm$ 0.21 | <b>3.44</b> $\pm$ 0.06 | <b>4.30</b> $\pm$ 0.06 | <b>3.57</b> $\pm$ 0.03 | <b>11.55</b> $\pm$ 0.06 |

GATE-KD achieves the highest retrieval average in configurations (b) and (c). In configuration (a), it ranks second to TALAS, with 16.12 versus 18.14 Avg. All distilled students remain substantially below their teachers, indicating that the sentence-level distillation objectives do not fully preserve the teachers’ zero-shot retrieval capability.



## E LAYERWISE REPRESENTATION CHANGES

We examine whether final-embedding supervision changes the internal representations of the student even though GATE-KD applies no intermediate-layer objective. The probe contains 512 unique sentences sampled from the validation splits after excluding sentences found in the distillation corpus. For every teacher and student Transformer layer, we compute centered linear CKA between their pooled representations (Kornblith et al., 2019). Results for the trained students are averaged over seeds 42, 43, and 44. We omit the embedding-output layer because special-token pooling makes this representation constant across examples, for which centered CKA is undefined.

Figure 2 shows that the pretrained students already share substantial representation structure with their teachers. Raw CKA alone therefore combines correspondence inherited from pretraining with changes induced by distillation.

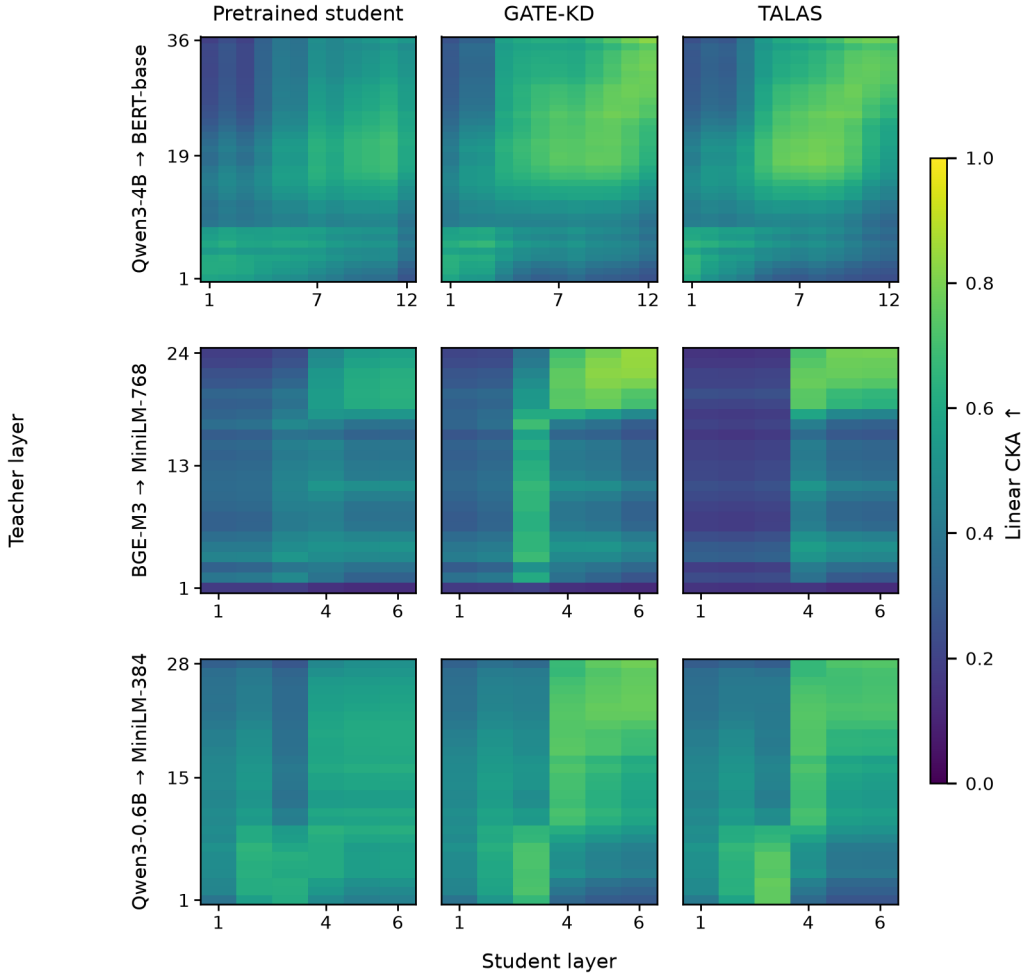


Figure 2: Layerwise linear CKA for the pretrained student, GATE-KD, and TALAS across three configurations. Trained results are averaged over three seeds.

To isolate the training-induced change, Figure 3 subtracts the pretrained-student CKA matrix from the mean matrix after distillation. GATE-KD produces higher final-to-final CKA than TALAS in all three configurations. The raw values are 0.8266, 0.8447, and 0.7845 for GATE-KD, compared with 0.7741, 0.7901, and 0.7333 for TALAS in configurations (a), (b), and (c), respectively. The corresponding changes from initialization are 0.2992, 0.2773, and 0.2781 for GATE-KD, compared

with 0.2468, 0.2227, and 0.2269 for TALAS. The final-layer advantage is consequently consistent across the three configurations and ranges from 0.051 to 0.055 CKA.

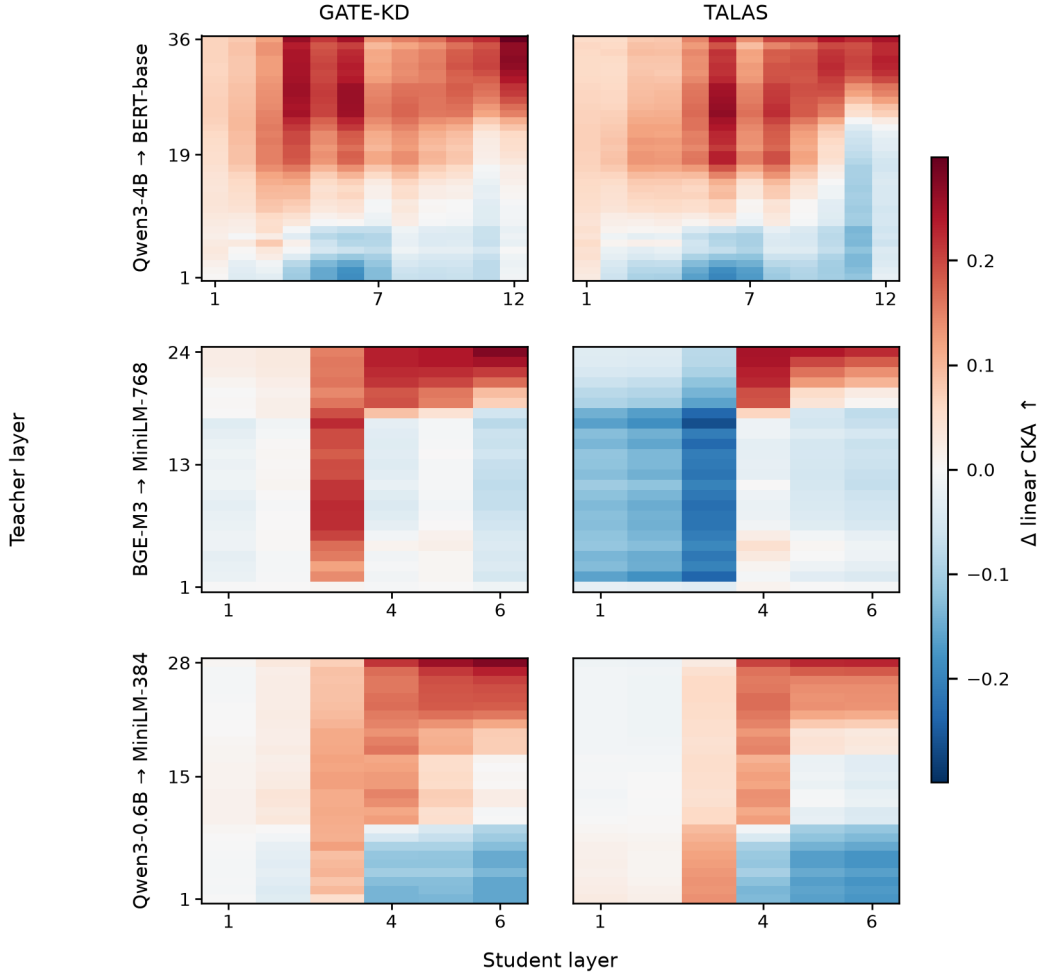


Figure 3: Change in layerwise linear CKA relative to the pretrained student. Red and blue denote increased and decreased similarity, respectively.

The changes are distributed across multiple later student layers rather than being confined to a single teacher–student layer pair. This pattern is consistent with endpoint supervision inducing an internal reorganization of the student despite the absence of explicit intermediate-layer matching. Some layer pairs decrease in CKA, so the result does not imply uniformly stronger alignment at every depth. Moreover, CKA is descriptive and does not establish that the observed internal changes cause the downstream improvements.

## F TARGET-RANK SENSITIVITY AT FIXED STUDENT WIDTH

We isolate the subspace selected by the endpoint interface in configuration (c). The student architecture, 384-dimensional output, distillation corpus, optimizer, training schedule, and Procrustes updates remain fixed, while the retained PCA rank varies over  $k \in \{64, 128, 256, 384\}$ . This experiment uses the endpoint-only objective. Targets with  $k < 384$  are embedded in the native student space by appending zero coordinates before Procrustes alignment. This padding preserves their Gram matrix and active rank, and does not change the capacity or width of the student.

Retained energy is measured on the complete teacher cache. On a fixed sample of 2,048 normalized teacher embeddings  $T$ , we measure the normalized squared Gram distortion  $\|TT^\top - Y_k Y_k^\top\|_F^2 / \|TT^\top\|_F^2$ , where  $Y_k$  is the normalized rank- $k$  PCA target. We also report the normalized rank floor  $\sum_{j>k} \sigma_j(T)^4 / \sum_j \sigma_j(T)^4$  from Equation 7. The Gram quantities are independent of the orthogonal gauge and are therefore computed once, while downstream scores are averaged over three seeds.

Figure 4 shows monotonic changes in all three quantities. Increasing  $k$  from 64 to 384 raises retained teacher energy from 59.32% to 92.84%, reduces normalized Gram distortion from 6.797% to 0.086%, and improves Avg. from 72.57 to 74.27. The intermediate ranks 128 and 256 attain 73.77 and 74.21 Avg., respectively. Most of the downstream benefit is therefore recovered by rank 256, with the final increase to rank 384 contributing only 0.06 points in this configuration.

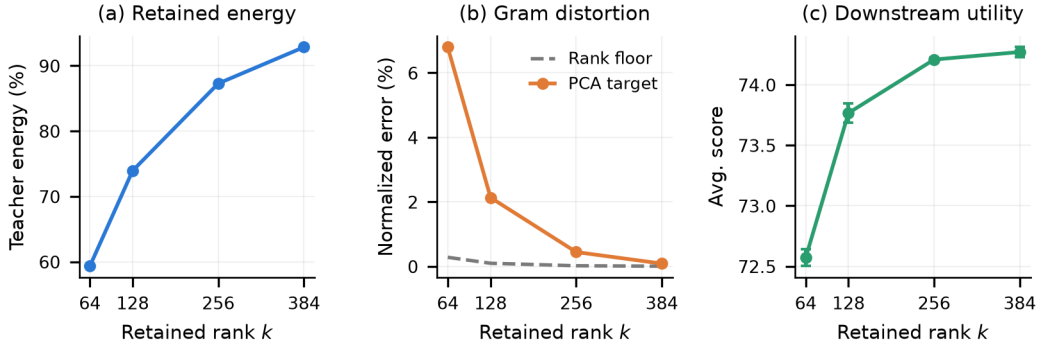


Figure 4: Sensitivity to retained PCA rank at fixed student width in configuration (c). Downstream scores show mean  $\pm$  sample standard deviation over three seeds.

The observed PCA distortion remains roughly 21–25 times above the rank floor. This difference is expected because the bound applies to the optimal rank- $k$  Gram approximation, whereas PCA maximizes retained variance before row normalization and does not directly optimize the normalized Gram matrix. The floor certifies irreducible error rather than predicting the distortion of the chosen interface. Overall, the sweep links the retained teacher subspace to both geometric fidelity and downstream utility, while its saturation indicates diminishing returns from additional principal directions. Because the student architecture remains fixed, this result is a target-rank sensitivity analysis rather than an ablation of student width.